

Generation of High-Power, Tunable Terahertz Radiation from Laser Interaction with a Relativistic Electron Beam

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May 4, 2018

IPAC'18, Vancouver, Canada

☐ Introduction

- Demands for THz radiation
- Accelerator-based methods for THz generation

☐ Proposed THz source

- Theory
- Simulation
- Radiation generation

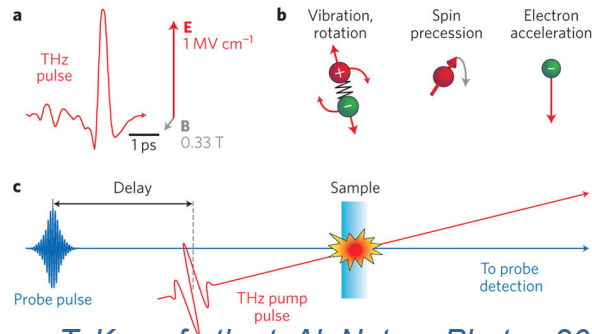
☐ Recent and proposed experimental studies

☐ Discussions and summary

THz science

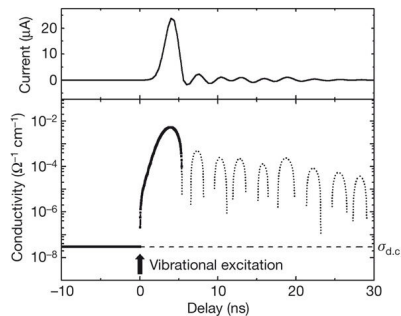
- THz science is a rich field with the potential to advance research in many scientific areas.

Nonlinear (High-field) THz experiment



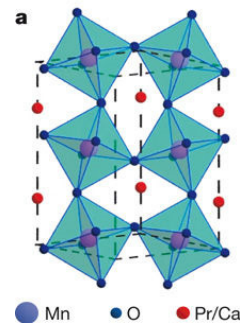
T. Kampfrath et al. Nature Photon 2013

THz control of conductivity

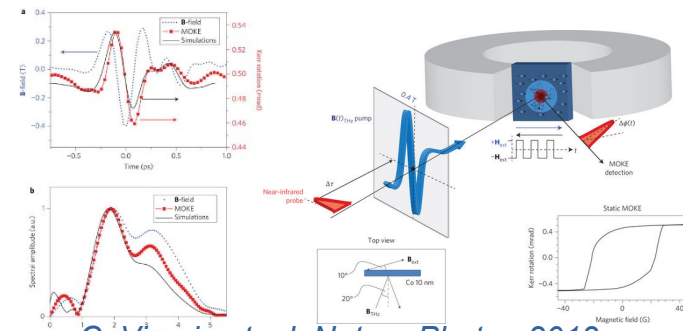


M. Rini et al. Nature 2007

M. Liu et al. Nature 2012



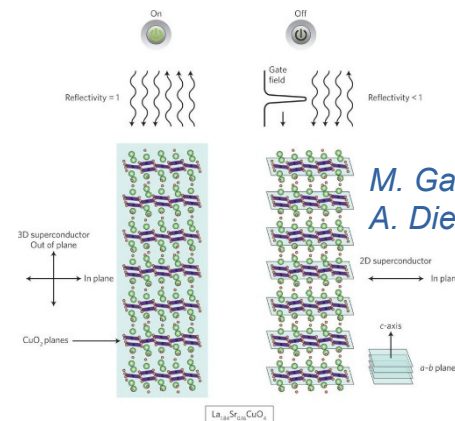
THz control of magnetism



C. Vicario et al. Nature Photon 2013

T. Kampfrath et al. Nature Photon 2011

THz control of superconductivity



M. Gaby et al. Nature Photon 2011

A. Dienst et al., Nature Materials 2011

Continued interests in the community

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- Very strong interests exist in combining THz radiation with X-ray FELs for pump-probe experiments



European XFEL Workshop

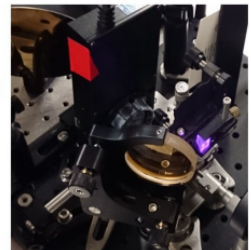
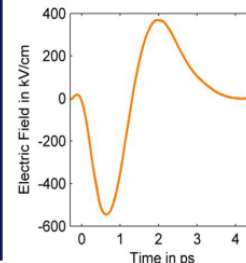
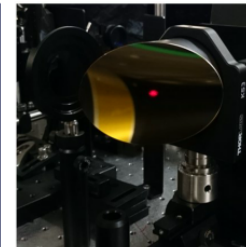
Terahertz science at European XFEL



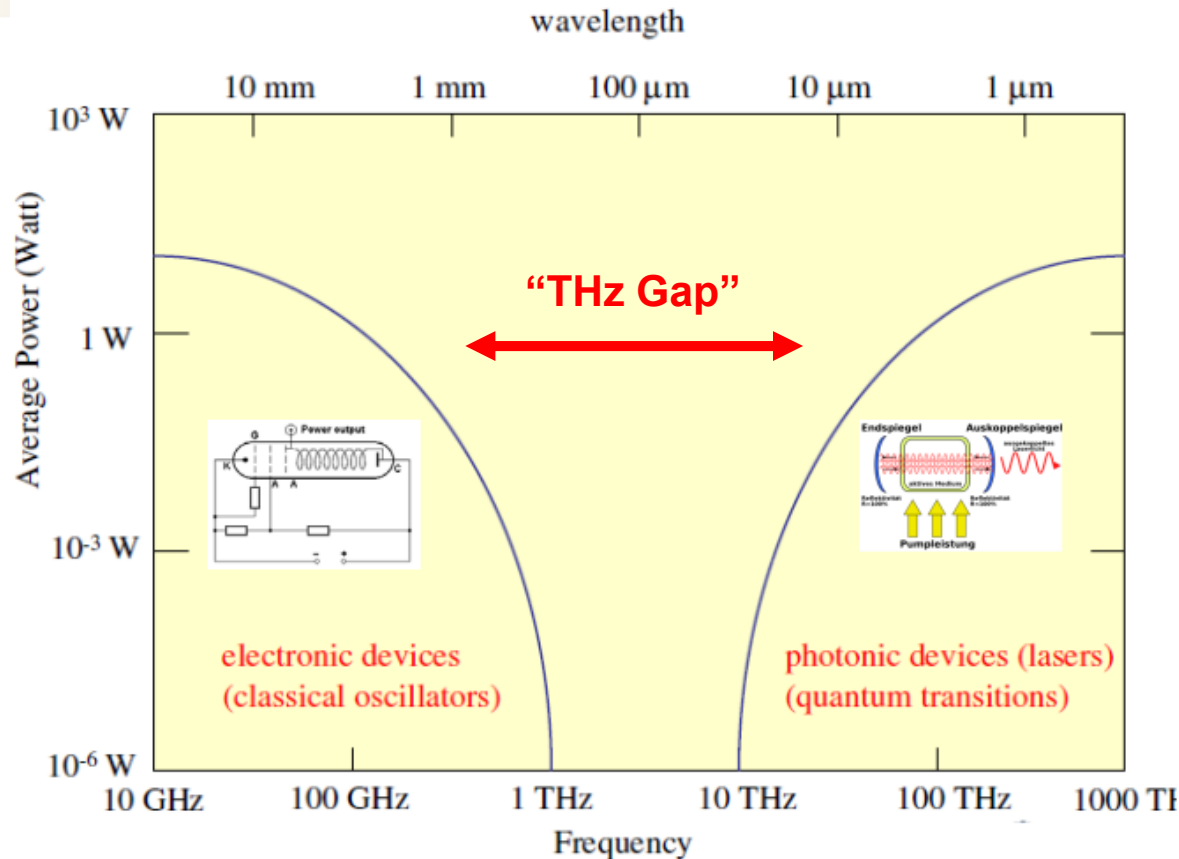
01–02 June 2017 / European XFEL, Schenefeld, Germany

The **interaction of terahertz radiation with matter** excites novel states with unique properties. These non-equilibrium states will benefit from ultrafast characterization using the ultrashort and highly brilliant X-ray pulses available at FELs. With this in mind, we are pleased to announce a terahertz workshop, with the aim to find the most scientifically promising strategies to combine terahertz radiation with the unique X-ray pulses generated at European XFEL.

The 2 day workshop will be led by 18 presentations from invited international scientists. The first day will explore the **scientific motivation for THz - X-ray experiments**. The second day will focus on the **two main routes to THz generation in the frequency range from 0.1 to 20 THz** (3 mm to 15 μm): On the one hand, state of the art laser-based sources; and on the other, undulator sources based on a second, smaller accelerator. Specific aspects will be compatibility with the MHz repetition rate of the European XFEL and novel opportunities for coherent control in the multi-THz regime. We intend to discuss how individual research projects can benefit from a combination of these sources.



THz gap (1 to 20 THz)



- Laser-based sources have made significant progress, but very challenging to reach above a few THz.
- Accelerator based sources are well suited for high-field, high-frequency, high-rep. rate THz applications.

Accelerator-based THz source

- Pre-bunched electron beam and coherent radiation
 - Convert transverse modulation to density bunching

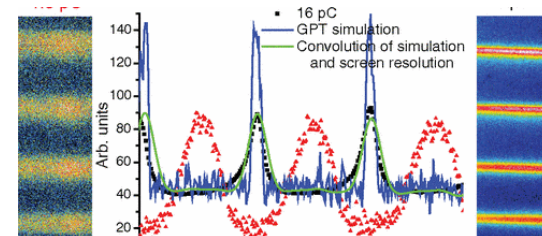
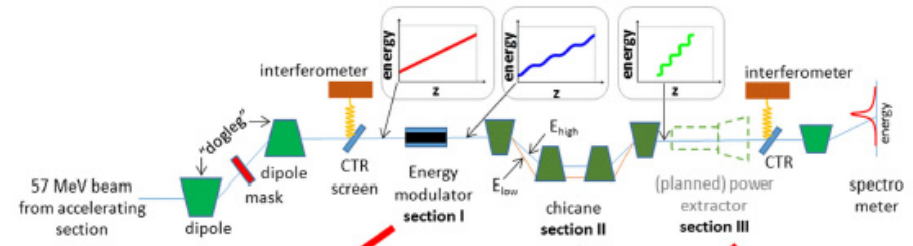
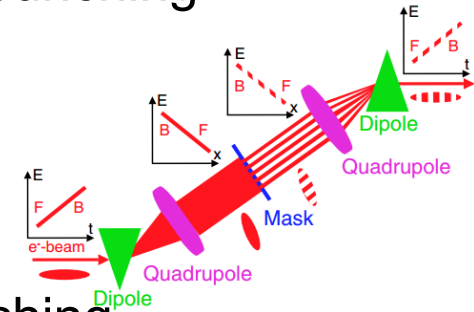
Transverse mask
+
Dogleg / Emittance exchange

- Convert energy modulation to density bunching

Wakefield / Laser interaction
+
Magnetic chicane

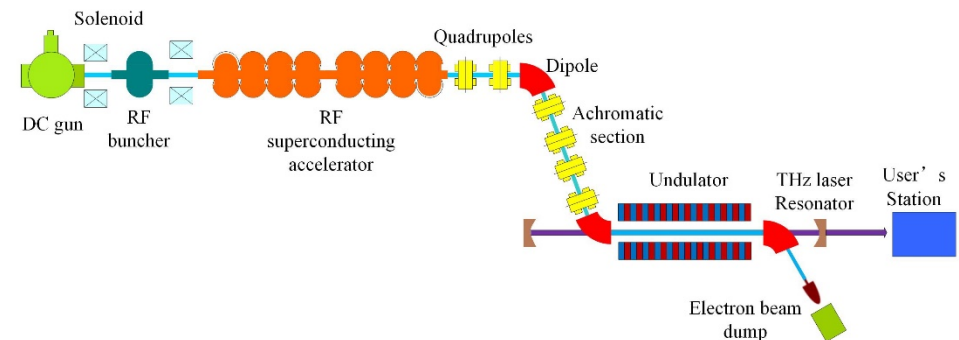
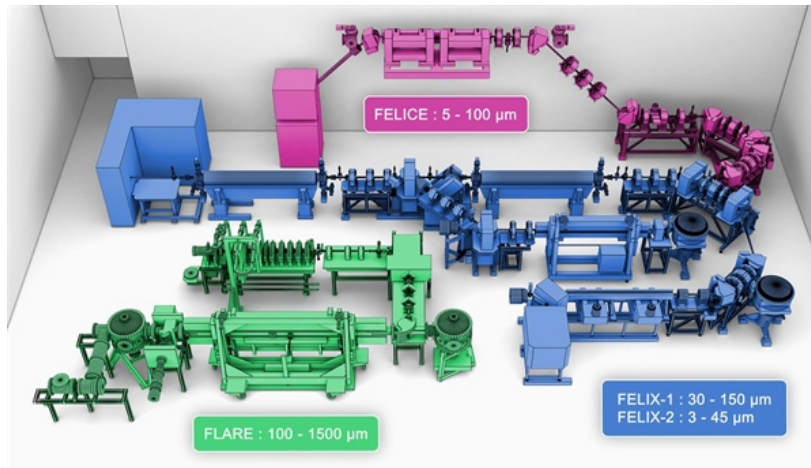
- Modulate drive laser to introduce initial density modulation

Laser shaping
+
Space charge oscillation



Accelerator-based THz source

- High-rep. rate electron beam and THz FEL oscillator
 - FELIX at Radboud University, The Netherlands
 - THz-FEL at Osaka University, Japan
 - THz facility at ELBE, HZDR, Germany
 - CTFEL at China Academy of Engineering Physics, China

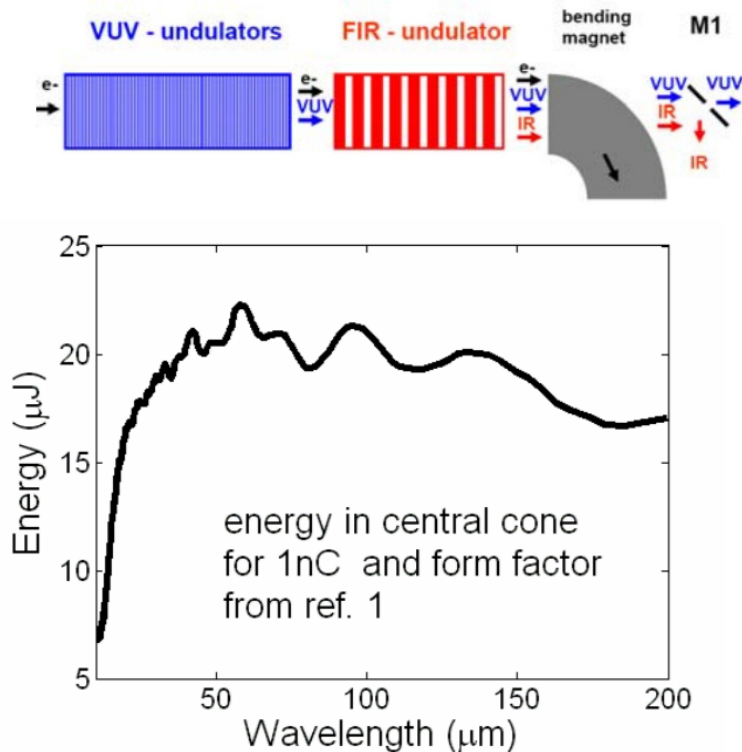


FELIX, Radboud University, The Netherlands

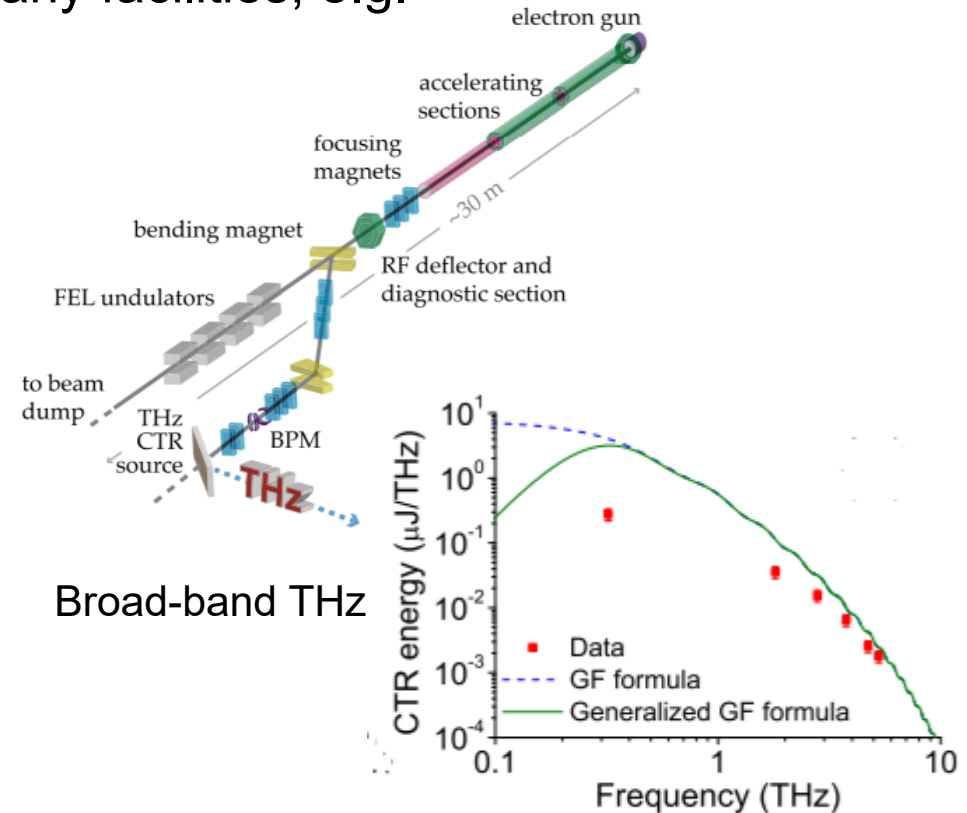
CTFEL, CAEP, China

Accelerator-based THz source

- Ultra-short electron beam and coherent transition/undulator radiation
- This method is widely used in many facilities, e.g.



FLASH afterburner, FIR undulator



SPARC linac, CTR

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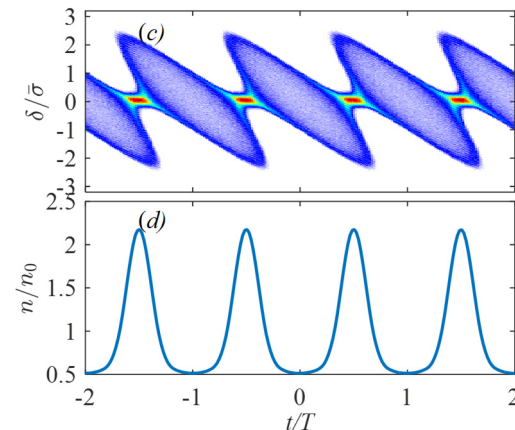
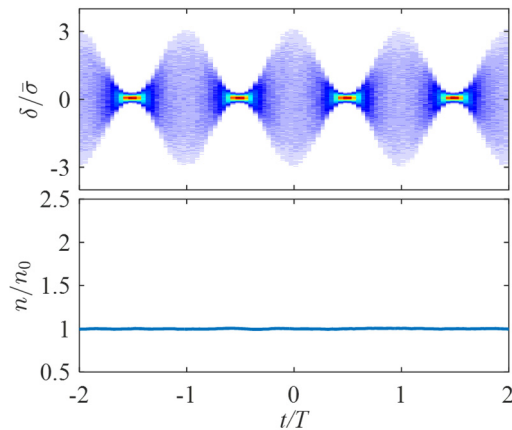
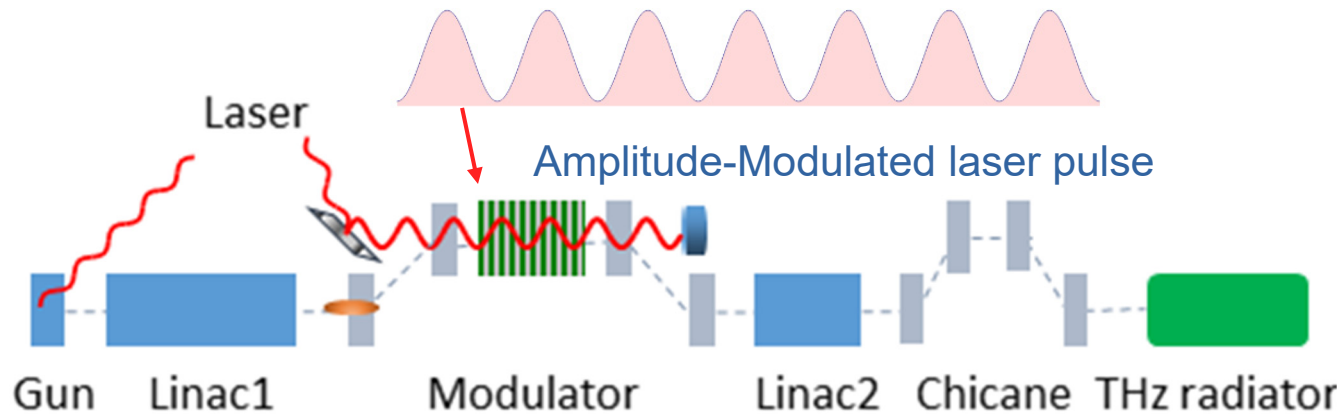
- Theory
- Simulation
- Radiation generation

□ Recent and proposed experimental studies

□ Discussions and summary

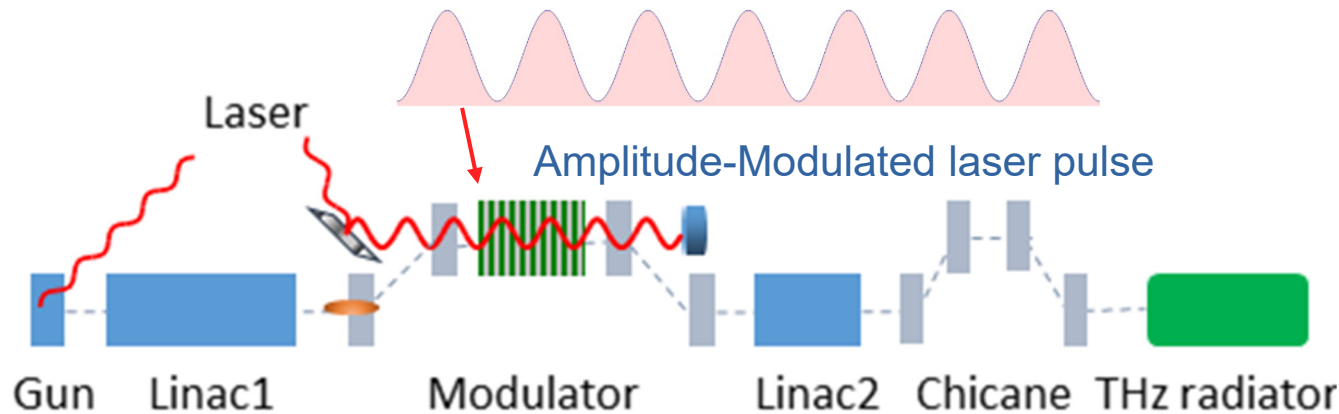
Slice energy spread modulation

- We propose a method based on the slice energy spread modulation to generate density bunching in a relativistic electron beam (a la **laser heater setup**)

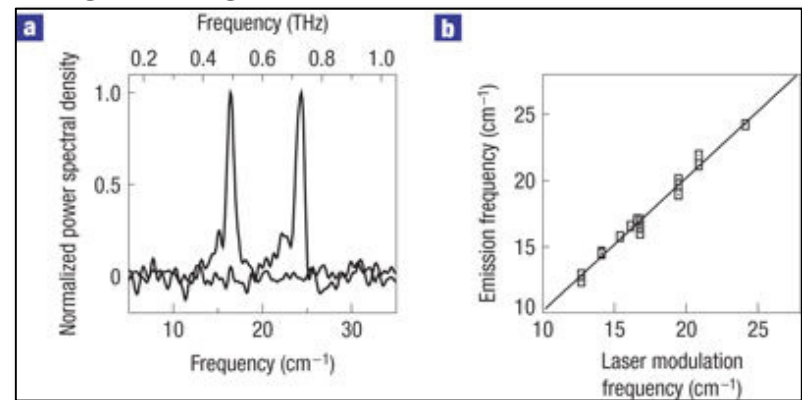
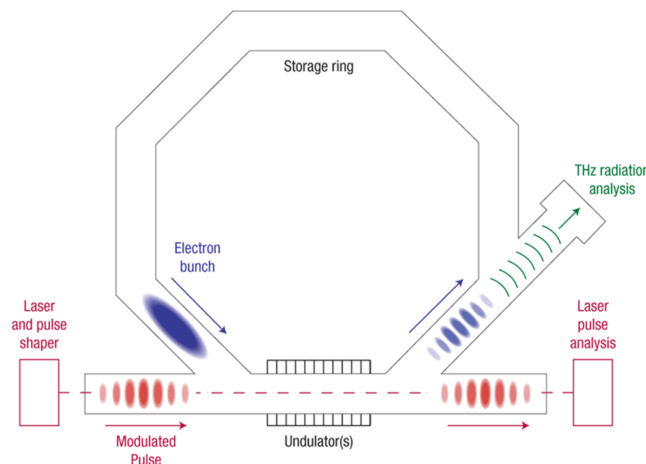


Slice energy spread modulation

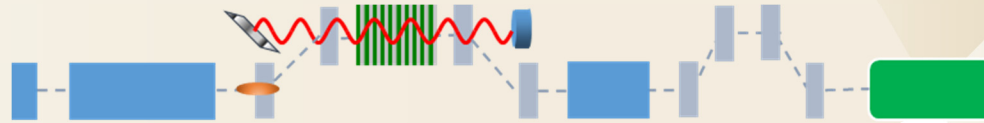
- We propose a method based on the slice energy spread modulation to generate density bunching in a relativistic electron beam (a la **laser heater setup**)



- Similar method has been used in storage ring THz generation



Theory



- Assume $f(\delta, z)$ is the beam distribution function. The bunching factor after the chicane can be written as

$$b(k) = \frac{1}{I_0} \int d\delta dz e^{-ikz} f(\delta, z)$$

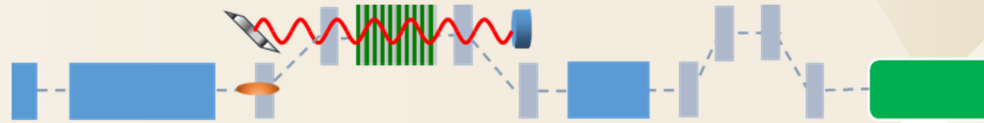
- Using the Liouville theorem $f(\delta, z) = f_0(\delta_0, z_0)$, $d\delta dz = d\delta_0 dz_0$, and making a change of variable from δ_0 to

$$\eta = \frac{\delta_0}{1 + A \sin(k_0 z_0)}$$

the bunching factor becomes

$$\begin{aligned} b(k) &= \int d\delta_0 dz_0 e^{-ik(1+hR_{56})z_0 - ikR_{56}\delta_0} \frac{1}{\sqrt{2\pi}\sigma_\delta(z_0)} \exp\left[-\frac{\delta^2}{2\sigma_\delta(z)^2}\right] \\ &= \int d\eta dz_0 e^{-ik(1+hR_{56})z_0 - ikR_{56}\eta[1+A \sin(k_0 z_0)]} \frac{1}{\sqrt{2\pi}\bar{\sigma}} \exp\left[-\frac{\eta^2}{2\bar{\sigma}^2}\right] \\ &= \frac{1}{\sqrt{2\pi}\bar{\sigma}} \int d\eta \exp\left[-ikR_{56}\eta - \frac{\eta^2}{2\bar{\sigma}^2}\right] \int dz_0 e^{-ik(1+hR_{56})z_0} \sum_n J_n(kR_{56}\eta A) e^{-ink_0 z_0} . \end{aligned}$$

Theory



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- Integration over z_0 yields nonvanishing bunching at the wavenumber

$$k_n = \frac{nk_0}{1 + hR_{56}} = nCk_0 \quad C = \frac{1}{1 + hR_{56}}$$

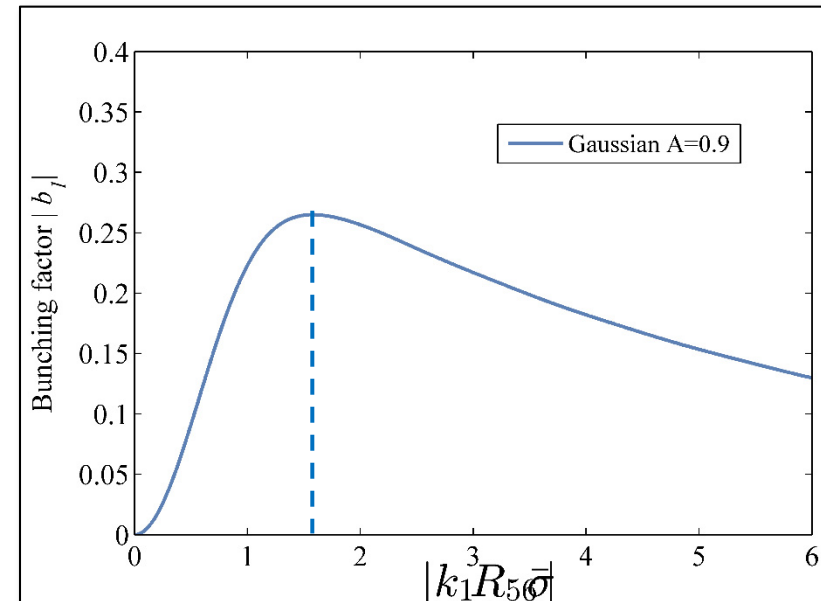
with the bunching factor

$$b_n = \frac{(-1)^n}{\sqrt{2\pi\bar{\sigma}}} \int d\eta J_n(k_n R_{56} A \eta) e^{-ik_n R_{56} \eta - \frac{\eta^2}{2\bar{\sigma}^2}}$$

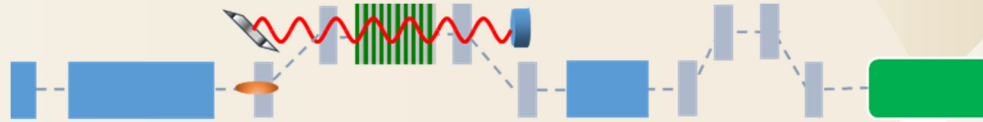
- Numerical calculation can be used to find the exact bunching factor and current distribution.
- The maximum bunching factor available at the fundamental wave number is ~ 0.27 .
- The optimal setting of chicane and laser pulse is

$$|k_1 R_{56} \bar{\sigma}| \approx 1.7$$

- $\bar{\sigma}$ is the energy spread after the laser interaction.

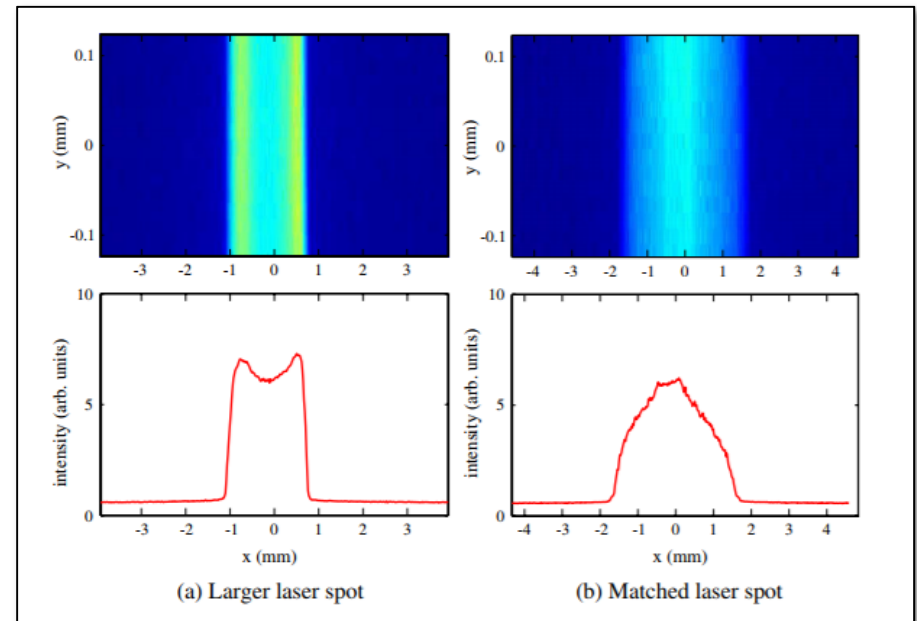
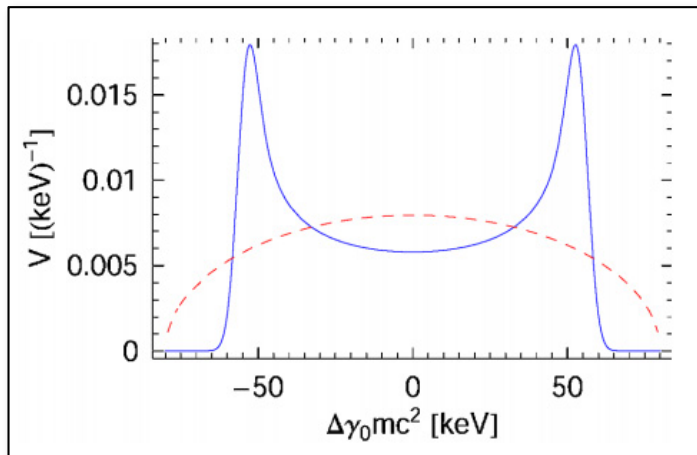


Theory



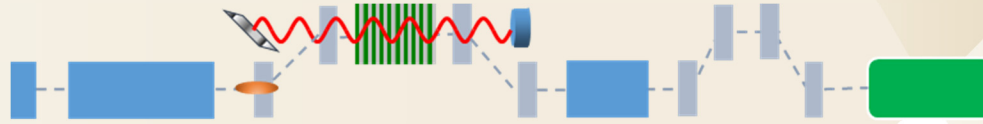
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- The analysis above assumes the beam has a Gaussian slice energy distribution, which is not always true in the laser heater.
- When the laser waist size in the undulator is much larger than the beam size, the resulting energy profile is a double-horn distribution



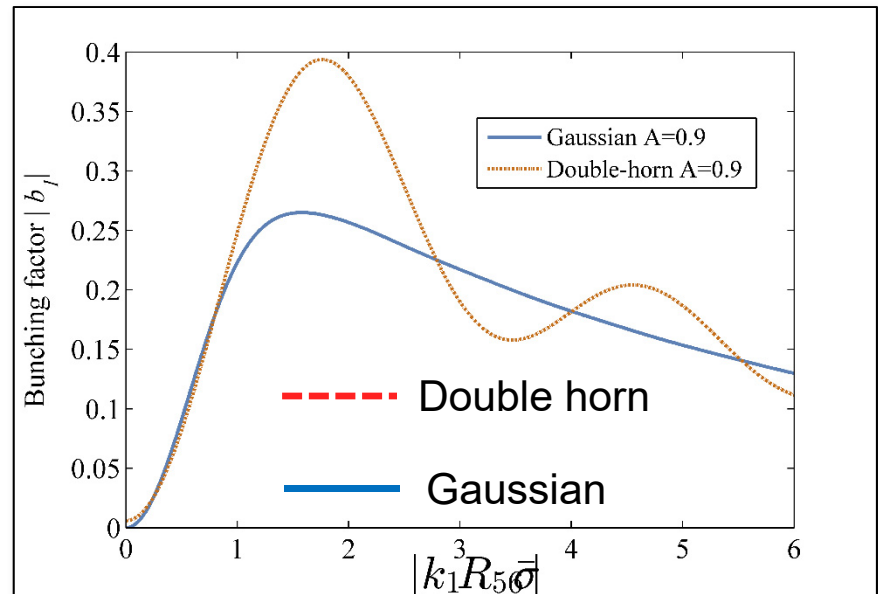
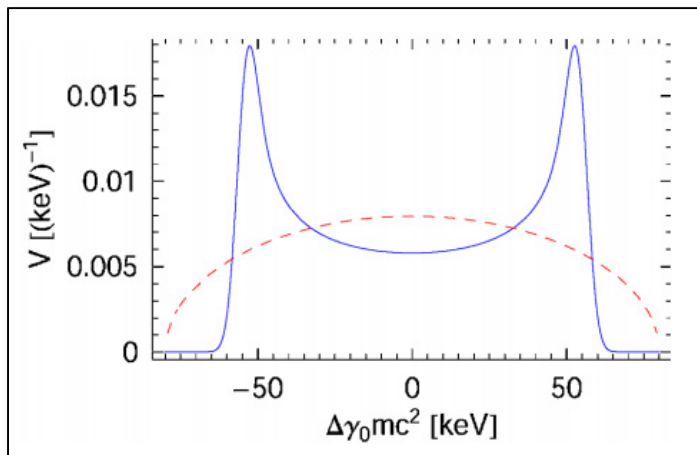
Z. Huang et al., PRST-AB 7, 074401 (2004)
Z. Huang et al., PRST-AB 13, 020703 (2010)

Theory



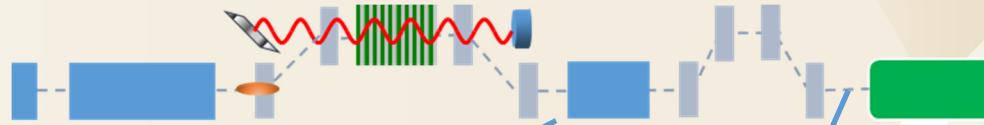
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- Replacing the Gaussian profile with the double-horn in the theoretical analysis, we find the double-horn profile is more effective to increase the bunching factor.
- Bunching factor of double-horn profile with the same modulation amplitude



- The maximum bunching factor is up to ~0.4!!

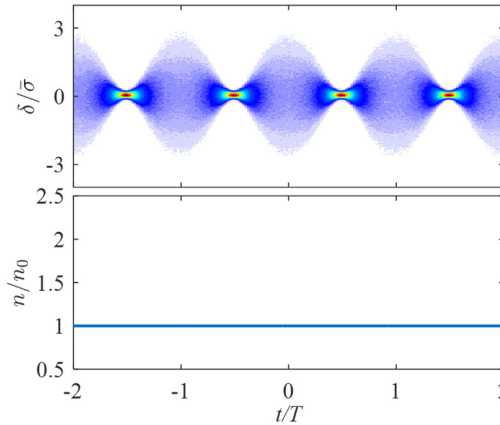
Theory



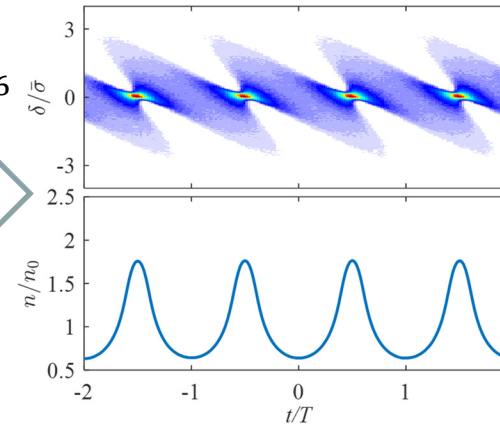
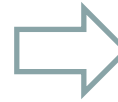
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- Phase space comparison

laser waist size
 $\gg$
 electron size

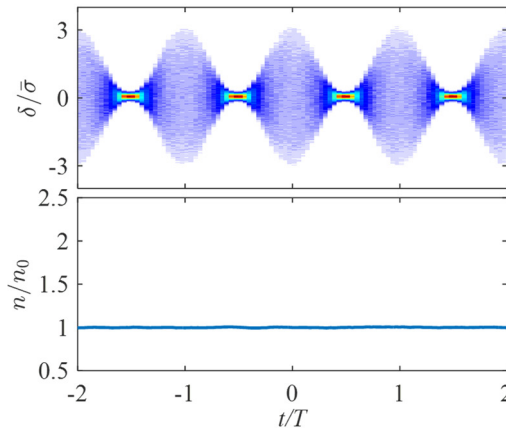


R_{56}

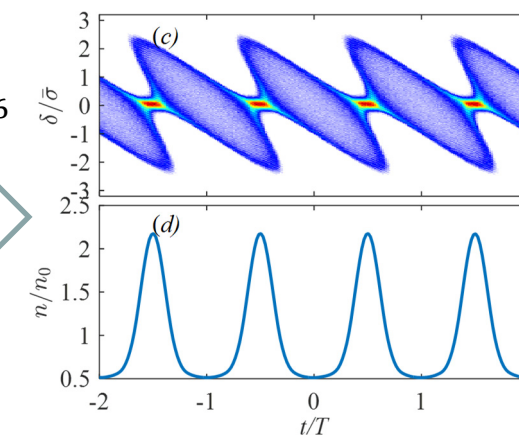
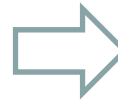


$b_1 \approx 0.27$

laser waist size
 $\gg$
 electron size
 Double-horn



R_{56}

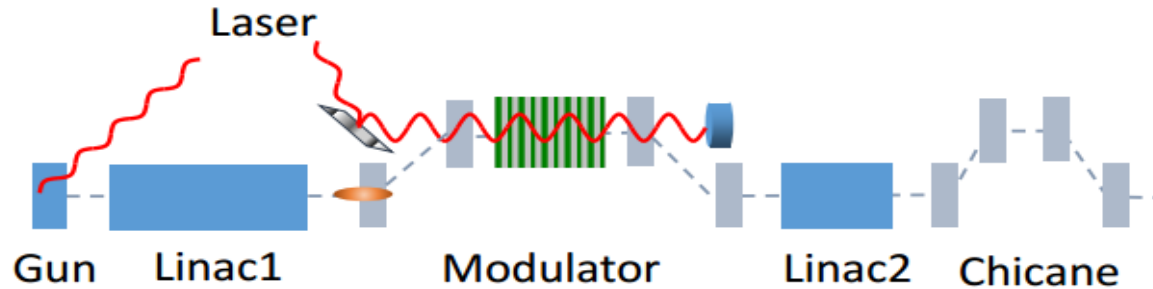


$b_1 \approx 0.4$

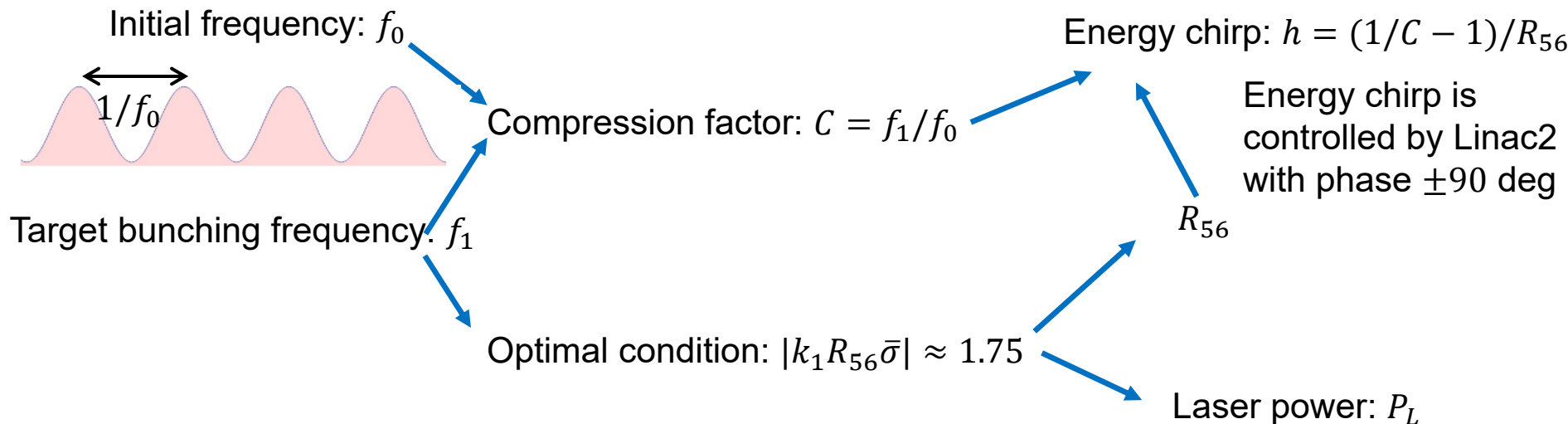
- Significant second harmonic bunching (~ 0.25) can be used to reach higher THz frequencies.

Simulation setup

- We use *ELEGANT* to simulate the laser modulation and electron bunching

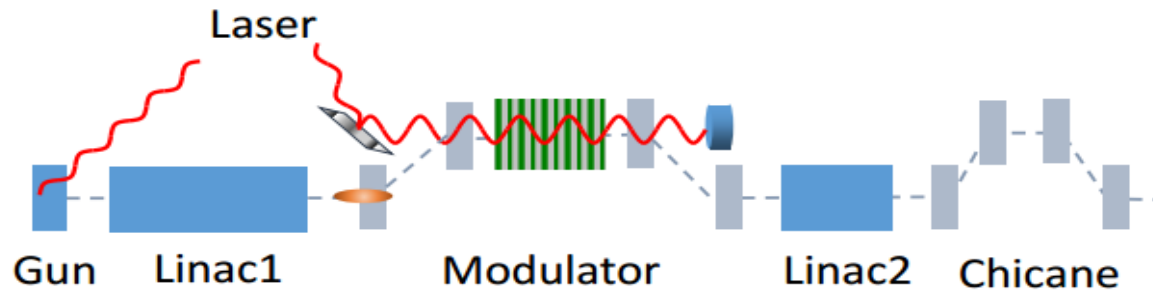


- The final bunching frequency is determined by initial modulation frequency and beam compression factor.



Simulation parameters

- Simulation parameters (LCLS injector)



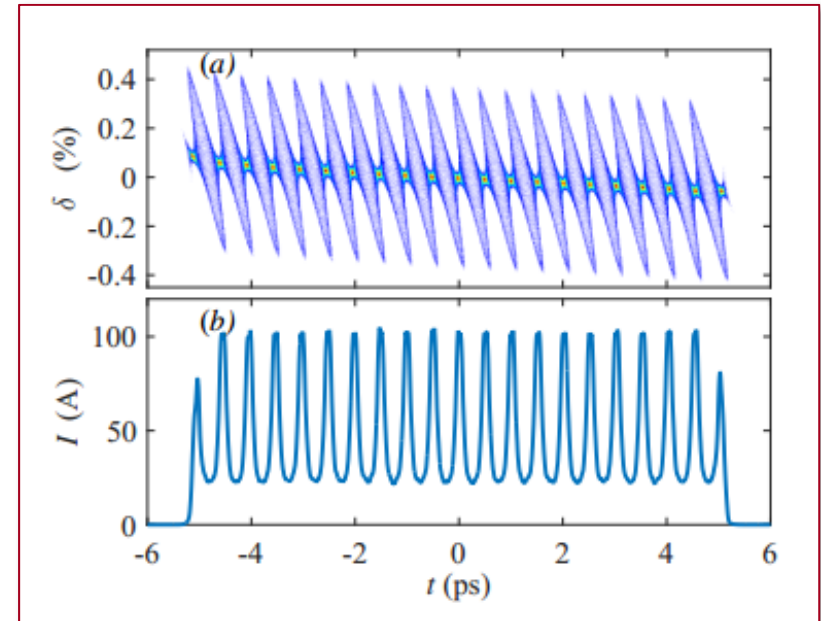
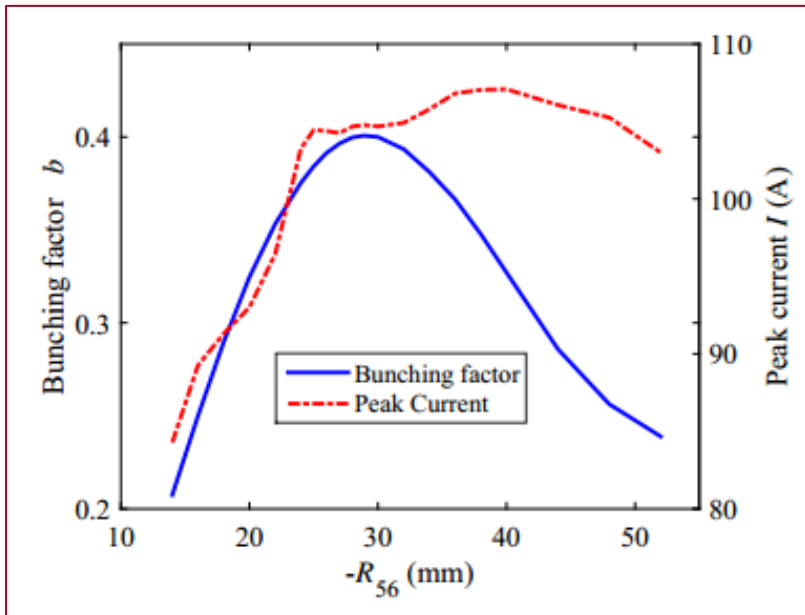
Parameter	Value	Units
Electron beam		
Charge	500	pC
Beam energy	135	MeV
Current profile	flatop	...
Bunch length	~10	ps
Intrinsic slice energy spread	10^{-4}	...
Normalized emittance	1	mm-mrad
rms beam size	200	μm

Modulator		
Laser wavelength	800	nm
Undulator period	5	cm
Period number	10	...
Laser waist size	1.5	mm
Laser stacking separation	0.5 (0.25 ^a)	ps
rms laser pulse length	60 (30)	fs
Laser power	1 (0.26)	GW

^aThe numbers in brackets are the parameters for the 4 THz case.

$f_1 = f_0 = 2\text{THz}$, no compression

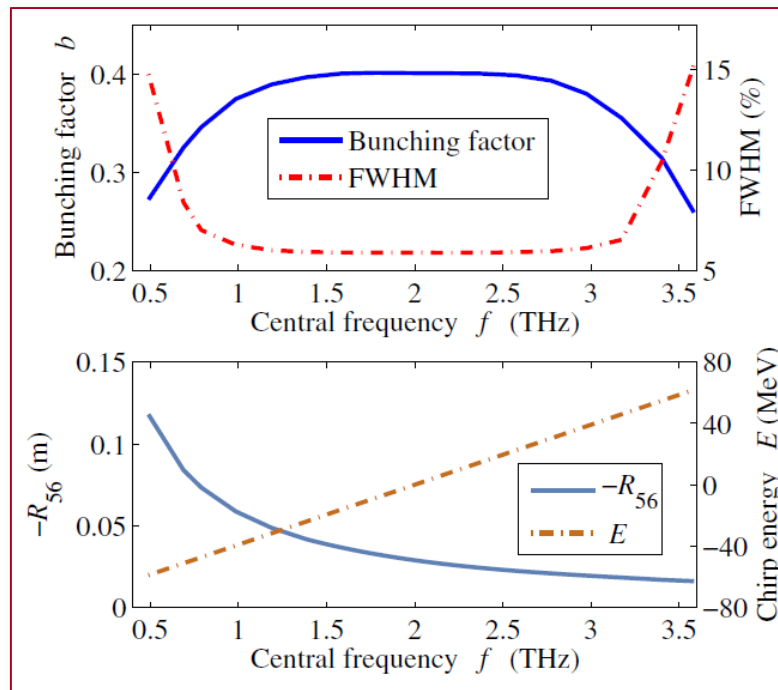
- Initial laser modulation frequency $f_0 = 2\text{THz}$, no energy chirp from Linac2 scan the R56 of the chicane with $\bar{\sigma} = 190\text{keV}$ ($P_L = 1\text{GW}$)



- The optimal condition $|k_1 \bar{\sigma} R_{56}| \approx 1.75$ predicts the best $-R_{56}$ is 29.4mm, consisting with the simulations (29mm).
- The longitudinal phase space and current profile when the $R_{56} = -29\text{mm}$ (optimal bunching)

$f_0 = 2\text{THz}$, frequency shift

- The bunching frequency of the e-beam can be varied by beam compression.
- As the change of bunching frequency, the product of R_{56} and $\bar{\sigma}$ needs to be varied to keep the optimal condition $|k_1 \bar{\sigma} R_{56}| \approx 1.75$
- Fix laser power (energy spread $\bar{\sigma}$), vary R_{56} and match energy chirp



Bunching factor and full-width-half-maximum of bunching spectra

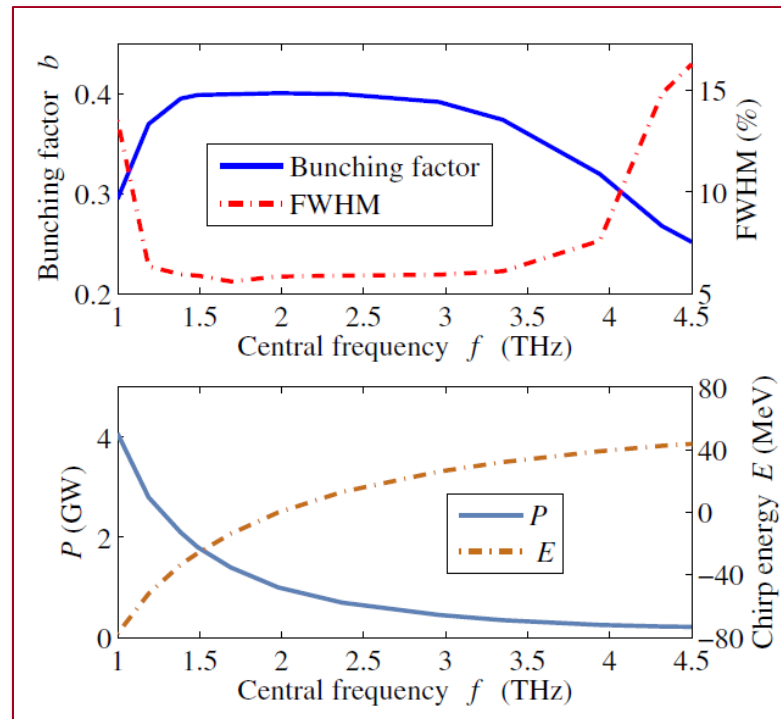


Required R_{56} and chirp energy in Linac2

- The decrease of the bunching factor is due to the non-uniform compression along the e-beam (nonlinear chirp and high-order terms in the chicane).

$f_0 = 2\text{THz}$, frequency shift

- The bunching frequency of the e-beam can be varied by beam compression.
- As the change of bunching frequency, the product of R_{56} and $\bar{\sigma}$ needs to be varied to keep the optimal condition $|k_1 \bar{\sigma} R_{56}| \approx 1.75$
- Fix R_{56} , vary laser power and match energy chirp



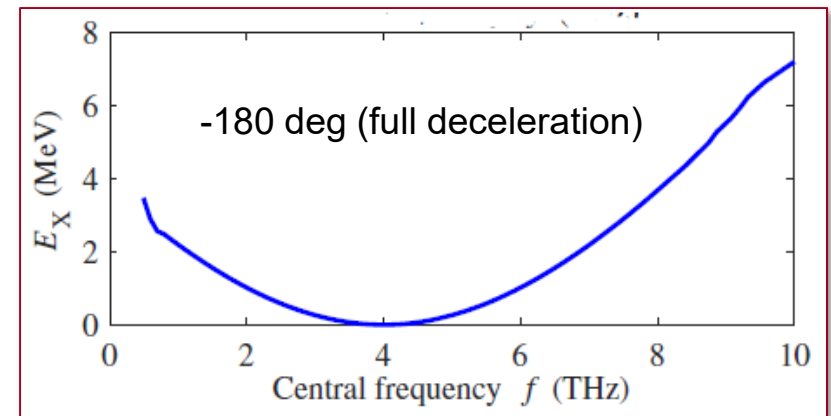
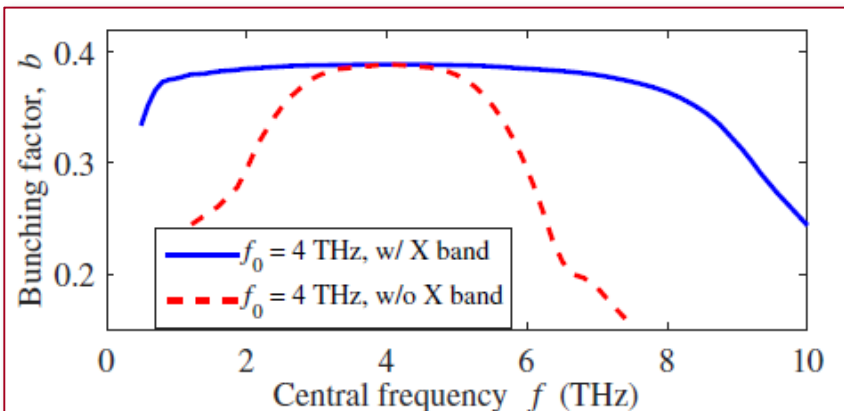
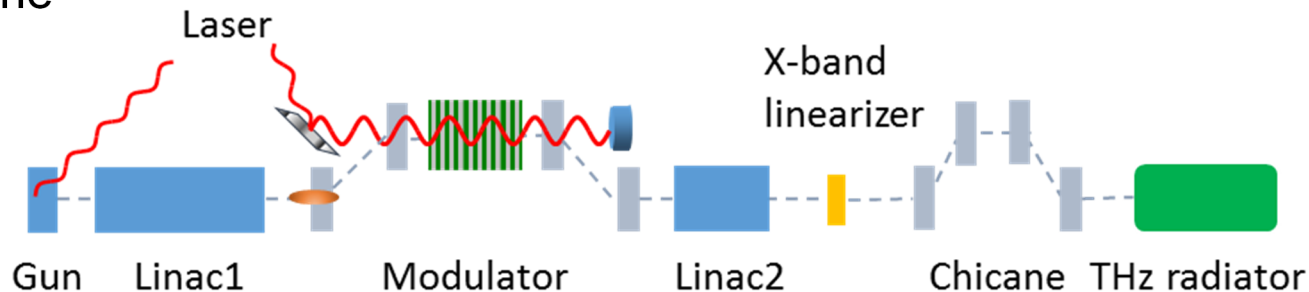
Bunching factor and full-width-half-maximum of bunching spectra



Laser power and chirp energy in Linac2

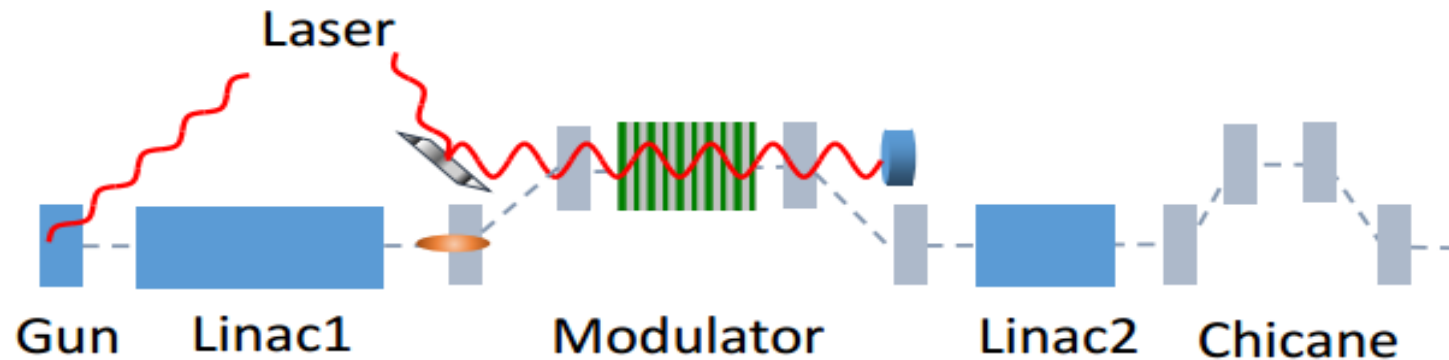
Larger frequency range

- Methods: higher initial modulation frequency and linearize energy chirp for larger beam compression
- E.g. $f_0 = 4\text{THz}$, and linearize energy chirp by high-harmonic cavity before chicane



Lower beam energy requirement

- The e-beam energy (135MeV in previous simulations) is limited by the resonant condition with 800nm laser.
- The energy requirement can be reduced if we choose high-harmonic resonance.



- Resonant wavelength in the undulator $2.4\mu\text{m}$
- Laser wavelength for modulation 800nm (3rd harmonic)
- E-beam energy $\sim 50\text{MeV}$, which is more attractive for THz generation

Compact accelerator-based THz source

- Compact accelerator of ~50 MeV interacts with a 800-nm laser in the undulator
- Laser-electron interaction through 3rd harmonic (for a planar undulator with fundamental resonant wavelength 2.4 μm .)

Parameter	Value
E-beam charge	1 nC
Beam energy	50 MeV
RMS beam size	0.2 mm
Bunch length (flat-top)	10 ps
Modulator	
Undulator period	2.5 cm
Peak field/ K value	0.56 T / 1.29
Undulator length/period	0.25 m / 10
Laser wavelength	800 nm
Laser RMS spot size	0.5 mm
Laser stacking separation	0.5 ps (2 THz)
Laser peak power	100 MW

Resonant condition:

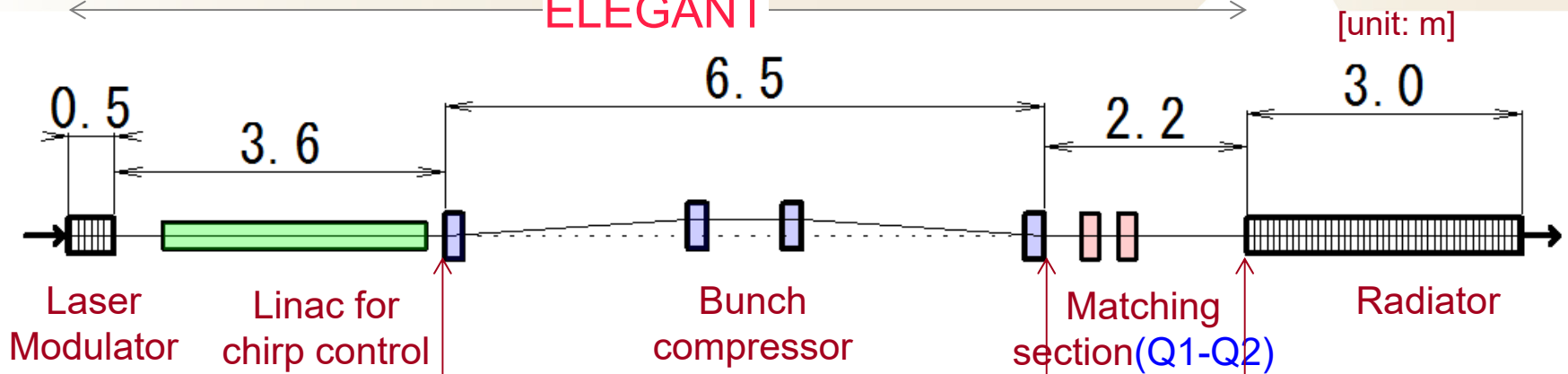
$$\lambda_r = \lambda_u \frac{1 + K^2/2}{2\gamma^2}$$

⇒ $\lambda_r = 2.4\mu\text{m}$

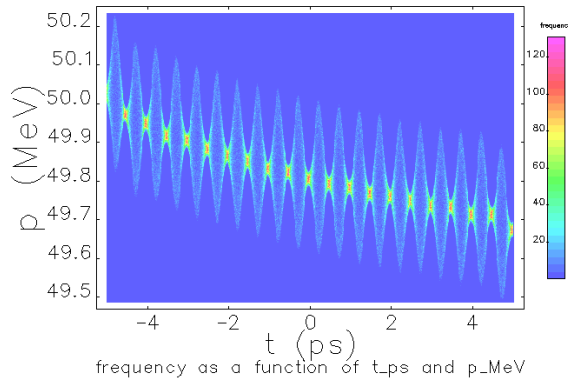
50 MeV simulation results

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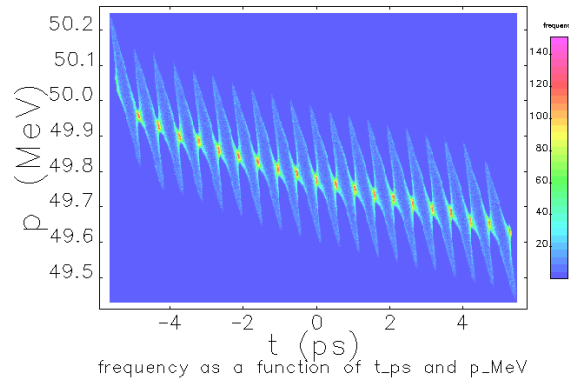


Data from SDDS file 03BC1BEG-t-ps-pMeV.h2d, table 1



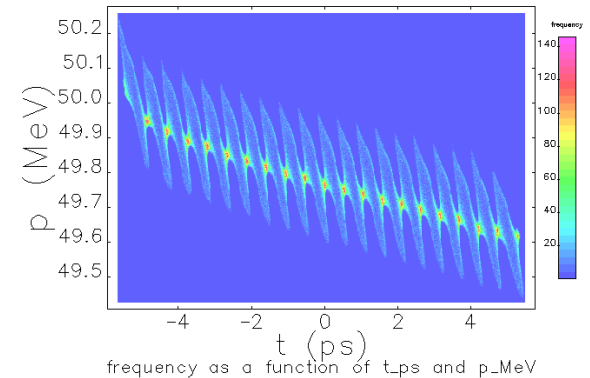
Before bunch chicane
(2 THz modulation
rms Espread=0.15%)

Data from SDDS file 04BC1END-t-ps-pMeV.h2d, table 1



After bunch
chicane
(bunching=0.37)

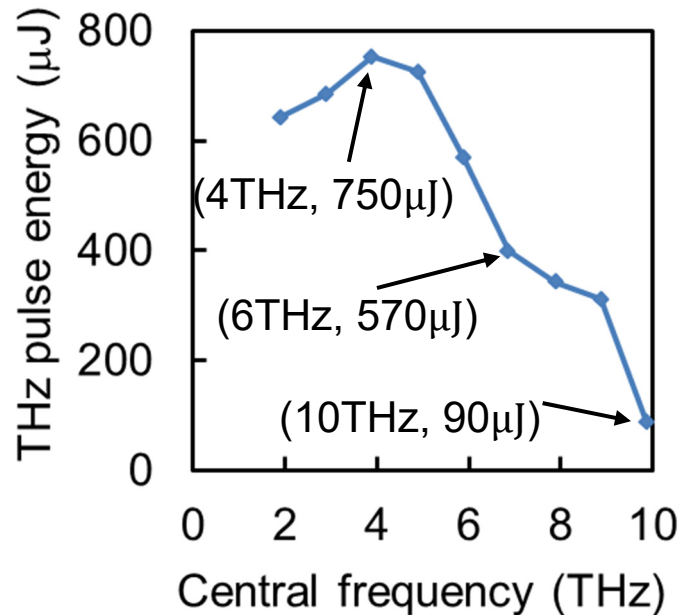
Data from SDDS file 05MatEND-t-ps-pMeV.h2d, table 1



After matching
section
(bunching=0.34)

THz radiation

- Transition radiation foil is the simplest radiator. CTR energy is $\sim 10 \mu\text{J}$ (5% bandwidth) with 1 nC charge.
- For a helical wiggler (radiator), the radiation energy at different central frequencies can be estimated by Genesis simulations.



$$\lambda_w = 10\text{cm} \quad N_w = 30$$

$$Q = 1\text{nC} \quad I = 100\text{A} \quad E = 50\text{MeV}$$

$$P_L = 100\text{MW}$$

Courtesy Koichi Kan

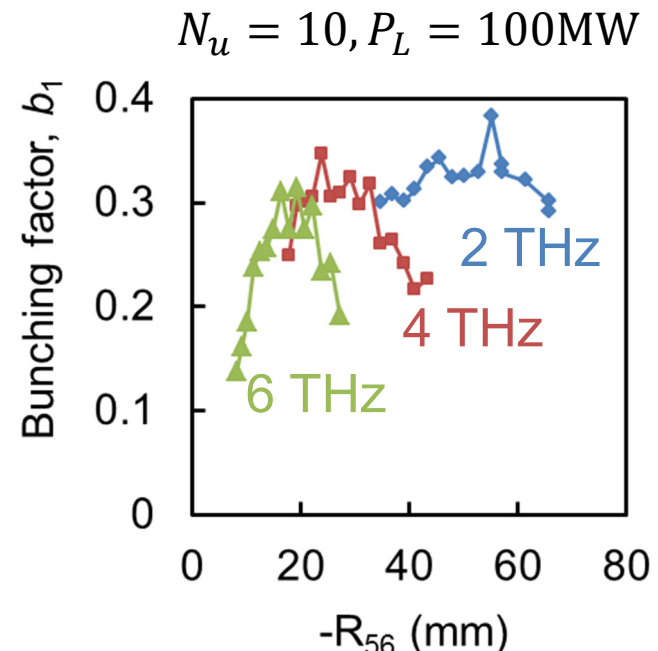
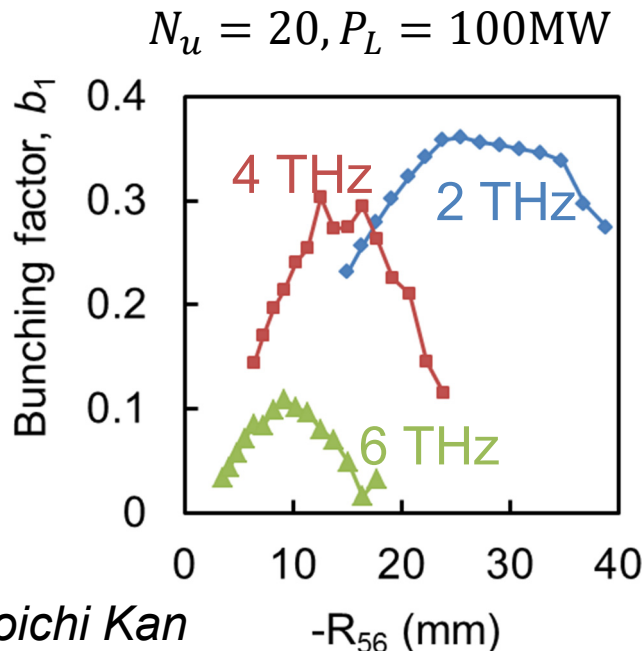
- Undulator in a waveguide or a dielectric tube may be more efficient radiators.

Slippage effect on bunching factor

- In the proposed scheme, the slippage in the modulator can smear the density bunching when it is comparable to the bunching wavelength.
- The slippage effect is more serious for high-harmonic resonance and high-frequency bunching.

$$\lambda_r = 2.4\mu\text{m}, N_u = 20, \text{Slippage} = 48\mu\text{m}$$

$$\lambda_{6THz} = 50\mu\text{m}, \lambda_{4THz} = 75\mu\text{m}, \lambda_{2THz} = 150\mu\text{m}$$



☐ Introduction

- Demands for THz radiation
- Accelerator-based methods for THz generation

☐ Proposed THz source

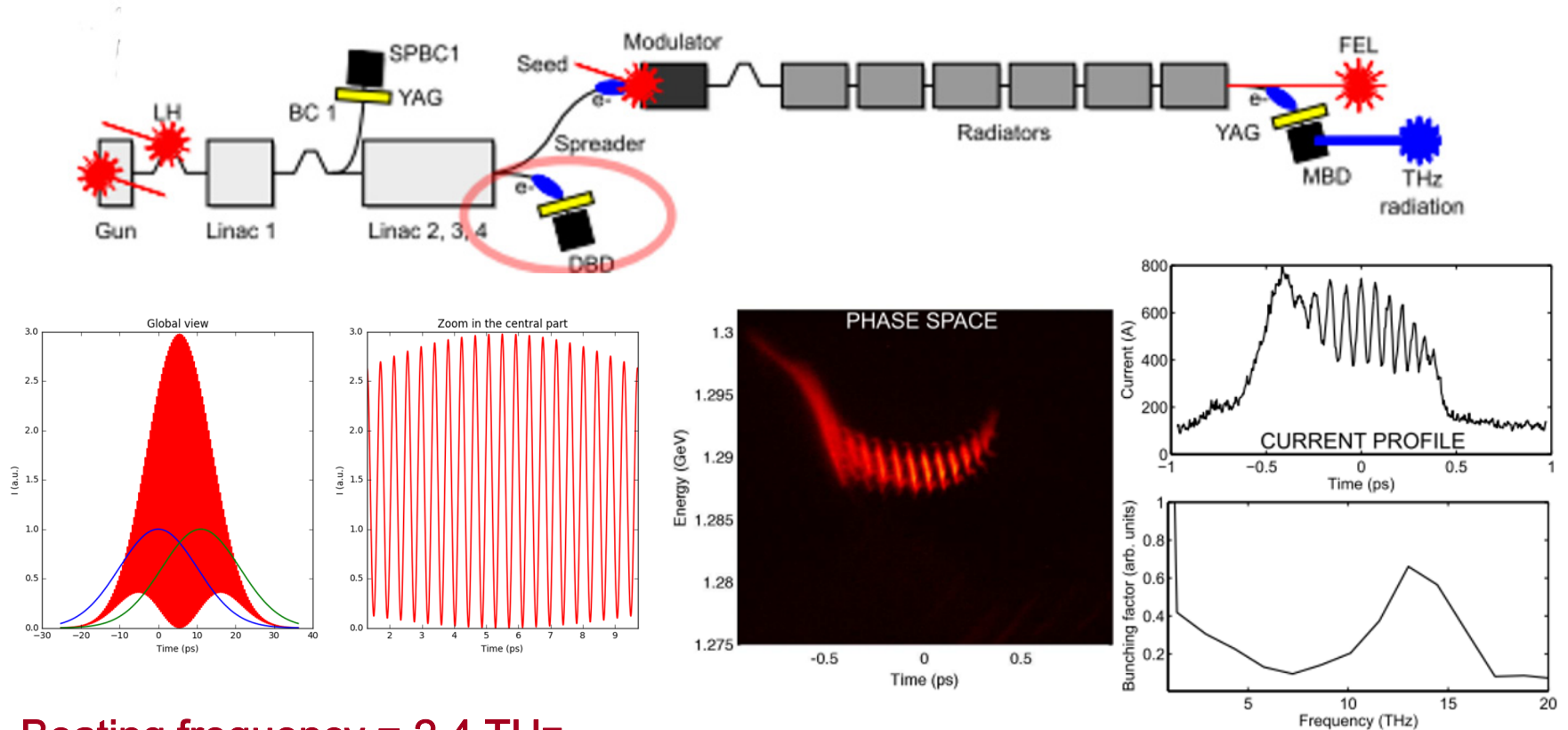
- Theory
- Simulation
- Radiation generation

☐ Recent and proposed experimental studies

☐ Discussions and summary

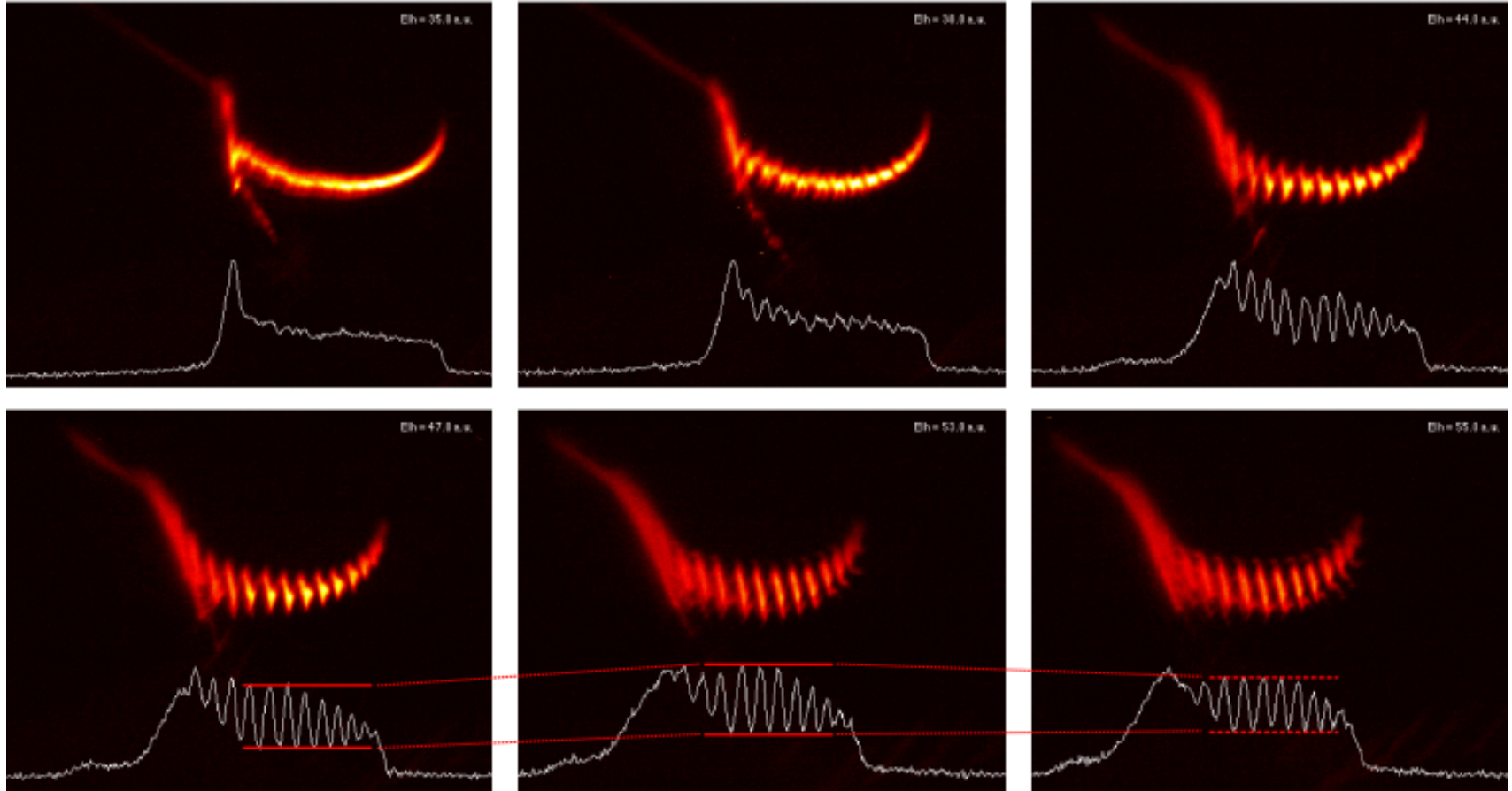
THz-modulated e-beam at FERMI

- Observation of a modulated beam in tens of THz range at the end of linac starting from a initial modulation in laser heater around 2.4THz.



THz-modulated e-beam at FERMI

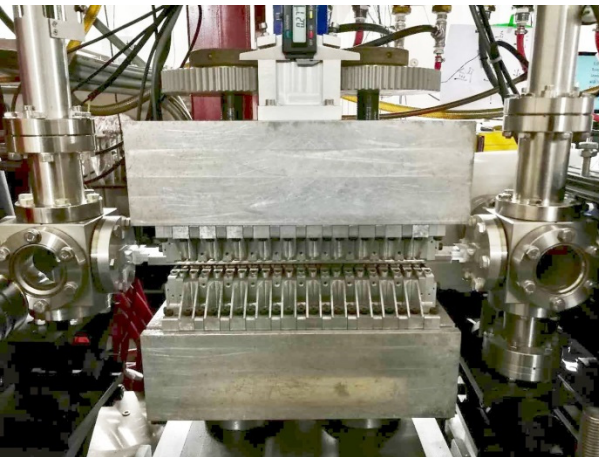
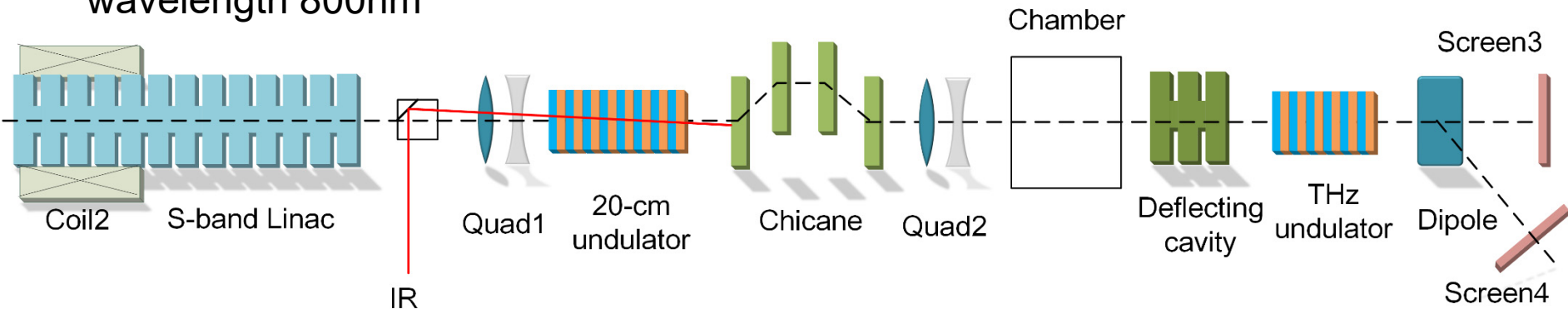
- By changing laser power (or by dispersion strength)



Proposed experiment at Tsinghua University

SLAC

- Beam energy $\sim 50\text{MeV}$, undulator period 2.5cm , $K = 1.3$, third harmonic resonant wavelength 800nm



Modulator
(works at 3rd harmonic)



chicane



THz undulator

- Introduction
- Proposed THz source
 - Theory
 - Simulation
 - Radiation generation
- Recent and proposed experimental studies
- Discussions and summary

- Based on the slice energy spread modulation method, the bunching factor can be kept around 0.4 for a wide frequency range (1-10 THz) and can be extended to 20 THz by compression or by taking advantage of the second harmonic bunching.
- THz pulse energy is estimated to be tens of μJ to hundreds of μJ .
- The method is also applicable for the electron beams from storage rings, ERL, or even thermal-cathode injectors with higher repetition rate.

Proposal of High-rep. Rate Stand-alone THz source

- Stand-alone THz source at the experimental area (XFEL, LCLS-II)
- Use LCLS-II spare gun + accel. cryomodule (50 MeV) for a high-rep. rate compact accelerator
- Leverage OPCPA laser at LCLS-II R&D (800 nm, 0.1 -1 MHz, 100 W)
- Expect good synchronization with hutch lasers (both through OPCPA)
- Strong THz field may be used in the LCLS(-II) TimeTool to cross-correlate with optical signals (and X-rays) for jitter corrections
- THz pulse form can be controlled by both laser and e-beam techniques (narrowband, chirped, a few cycle pulses, all possible)
- Flexible, powerful, high-rep. rate, well-synchronized with X-rays.

Acknowledgement

- Our Tsinghua University collaborators Lixin Yan, Yingchao Du, Wenhui Huang, and Chuanxiang Tang.
- Many useful discussions with E. Allaria, R. Coffee, B. Garcia, M. Hoffmann, A. Lindenberg, G. Marcus, A. Marinelli, K. Kan, E. Roussel, R. Schoenlein, F. Tavella,...

Thanks for your attention!